

MARK TRAVNIK

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EDUCATION

Boston University, College of Engineering

Boston, MA

Bachelor of Science in Mechanical Engineering

Expected May 2028

- Relevant Coursework: CAD & Machine Components, Circuits, Fluid Mechanics, Mechanics of Materials, Mechanics, Thermodynamics.

EXPERIENCE

Automation Technician Intern

Jun 2025 - Aug 2025

Navitas Systems Integration, Inc.

Plymouth, MA

- Modernized infrastructure using remote operated controls, resulting in frictionless end-user operation using Siemens software for efficient heating and cooling.
- Installed and wired HVAC components including Siemens valve actuators and DXR controllers.
- Balanced and commissioned HVAC control systems using Siemens ADT software to ensure accuracy of flow rates and temperature readings.
- Maintained efficient operation of HVAC components by programming, troubleshooting, and calibrating for accurate readings through Siemens ADT and Desigo CC software.
- Performed repairs on faulty systems across the Boston area, including flooded RTUs and failed cooling systems, informing facilities managers how to maintain stable and efficient operation.

PROJECTS

8 Servo Walker

Jul 2026 - Aug 2026

- Designed a walking quadrupedal robot capable of propelling itself forward utilizing an Arduino board commanding 8 individual servo motors.
- Programmed efficient C++ code actuating each motor according to precise angle and time data, resulting in a fluid and stable gait.
- Created component and wire diagrams to maximize space and efficiency and assembled components into a secure load-bearing body.

Truss Project

Sep 2025 - Nov 2025

- Built truss capable of holding 42+ oz of weight without buckling using material analysis.
- Developed complex MATLAB code capable of analyzing precise tension and compression forces on each joint, rendering a 2D image of each design allowing for efficient testing and troubleshooting.
- Collaborated with team to ensure project specifications were met including cost/load ratio, total material usage, and joint quantity.

Optimus Prime Senior Design Project

Jan 2024 - May 2024

- Designed transformable 3D model with multiple moving and interlocking parts using AutoCAD and Onshape, allowing for seamless SOLIDWORKS assembly and smooth movement in final print.
- Calibrated Ender-3 3D printer and used Cura slicing software to create an optimized print.
- Performed iterative prototyping and assembly with multiple materials ranging from PLA to PETG.

SKILLS

Software: MATLAB, Solidworks, Python, Java, JavaScript, C++, AutoCAD, Revit, Cura, Microsoft Office

Manufacturing: Soldering, Laser Cutting, 3D Printing, Circuit Assembly, Milling, Drill Press

Language: English, Russian

Certifications: Massachusetts Chapter 74 CCTE Drafting/CAD certificate awarded June 2024